ALVIN GEORGE

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Kochi, Kerala, India

Summary

Motivated B.Tech Computer Science student with hands-on experience in DevOps, Cloud Computing, Backend Development, and Infrastructure Automation. Skilled in building CI/CD pipelines, automating cloud deployments, developing scalable backend applications, and implementing monitoring and observability solutions. Passionate about designing reliable, secure, and production-ready software systems.

Work Experience

Backend Developer Intern – MarkX Jan 2026 – Mar 2026

- Developed backend services and REST APIs using FastAPI.
- Integrated AI-powered workflows using LangChain and LLMs (Using Groq).
- Built data collection and scraping pipelines using Apify.
- Contributed to cloud deployment and infrastructure-related tasks.
- Collaborated with frontend components built using React.js and deployed applications on Render.
- Tech Stack: Python, FastAPI, LangChain, Groq, Apify, React.js, Render.

Skills

Programming Languages	Java, Python, JavaScript
Backend Development	Spring Boot, Flask, FastAPI, Express.js, REST APIs
Frontend Development	React.js, Tailwind CSS
Databases	MongoDB, PostgreSQL, Neon DB
Cloud & DevOps	AWS, Google Cloud Run, Docker, Jenkins, Terraform, CI/CD
Monitoring & Observability	Prometheus, Grafana, PromQL
Tools	Git, GitHub, Pull Requests, VS Code, IntelliJ IDEA, Linux
Deployment Platforms	Render, Vercel, Firebase

Projects

Kerala Deforestation Detection System 2026–Present

- Built an end-to-end geospatial machine learning system to detect forest loss using Sentinel-2 satellite imagery and Google Earth Engine.
- Created a dataset of 50,000+ image patches across all 14 Kerala districts using Dynamic World land-cover labels and geographic train/validation/test splits.
- Trained an 8-channel U-Net (ResNet34 backbone) in PyTorch for pixel-level deforestation segmentation, achieving ˜0.36 validation Intersection Over Union (IoU) score.
- Developed and deployed a FastAPI inference service on Hugging Face Spaces, enabling real-time satellite image analysis from latitude/longitude inputs.
- Implemented preprocessing, cloud masking, patch generation, model inference, and visualization pipelines for remote sensing workflows.

DevOps Home Lab & Cloud Infrastructure — 2025

- Built end-to-end CI/CD pipelines using Jenkins, Docker, Sonatype Nexus, AWS EC2, and AWS ECR.
- Provisioned cloud infrastructure with Terraform, including EC2 instances, Security Groups, and remote state management.
- Deployed containerized applications on Kubernetes (Minikube) and explored self-healing and declarative infrastructure concepts.
- Implemented monitoring and observability using Prometheus, Grafana, and PromQL.
- Applied Linux administration practices including SSH hardening, Bash scripting, Nginx configuration, and secure server operations.

Digital Patient Card Platform — 2025

- Developed a multi-role healthcare management platform deployed on Google Cloud using Cloud Run and Artifact Registry.
- Built React-based dashboards for patients, doctors, and administrators using Tailwind CSS and React Router.
- Engineered a Spring Boot backend with REST APIs, JPA/Hibernate, secure authentication, and role-based access control.
- Integrated PDF generation using iText and implemented disease verification workflows.
- Automated deployment pipelines and integrated Neon PostgreSQL for scalable cloud-hosted storage.

Education

2024 – 2028	Bachelor of Technology in Computer Science Engineering at **Model Engineering College, Kochi**
2022 – 2024	Higher Secondary Education – Kerala State Board

Achievements

- Participated in HackForTomorrow Hackathon organized by EXCELMEC.
- Built an AI-based plagiarism detection system for websites.
- Secured Second Prize in the Tech Sprint Hackathon conducted by GDGC for the project *Huddle*.